Fiona Victoria Stanley Jothiraj

Beaverton, OR

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SUMMARY

EDUCATION

Oregon State University, Corvallis 2023 — 2027
Doctor of Philosophy (PhD) in Artificial Intelligence GPA: 3.79/4.00
Area of Research: AI for Social Good, Applied AI/ML in Ecology
Honors: Outstanding Scholars Program

University of Washington Bothell 2021 — 2023
Master of Science in Computer Science and Software Engineering GPA: 3.90/4.00
Thesis: Phoenix - A Federated Generative Diffusion Model

PSG College of Technology, India 2015 — 2019
Bachelor of Engineering in Robotics and Automation Engineering GPA: 9.58/10.00

RESEARCH EXPERIENCE

Graduate Researcher Oct 2023 - present
Oregon State University
Advisor: Dr. Rebecca Hutchinson

- Modeled plant–pollinator interactions across a decade of environmental change using Bayesian hierarchical dynamic occupancy models and probabilistic graphical models in R
- Investigated imperfect detection effects in species distribution models by simulating detection noise across linear and hierarchical statistical frameworks
- Conducted interdisciplinary research on species distribution modeling, demonstrating a 23% performance gain with boosted regression trees (BRTs) when habitat sampling is centered on bird locations rather than surveyor locations

AI Research Intern June 2024/25 - Sep 2024/25
Micron Technology
Advisor: Seth Eichmeyer, Director of Product Yield Analytics & Analysis

- Awarded the **Micron Innovation Award**, marking the first step toward a patent filing
- Designed and deployed ML models to optimize semiconductor product yield through wafer defect analysis
- Built a non-parametric clustering system with rotation- and flip-invariant features, reducing manual analysis time by 90% *(AAAI'26 Workshop & Bridge Program)*
- Developed a novel anomaly scoring metric using conditional VAEs, diffusion models, and ViTs to detect multi-defect patterns in semiconductor manufacturing
- Presented project outcomes and business impact to the VP of Engineering

Graduate Researcher June 2022 - Jan 2023
UW Bothell: Computation Behavioral Modeling (CBM) Research Lab
Advisor: Dr. Afra Mashhadi

- Conducted research on nostalgic behavior on social media using NLP and LLMs
- Built transformer models, including RoBERTa, DistilBERT, and an ensemble-based model for nostalgic tweet detection, achieving 0.96 Micro F1-Score
- Leveraged LLaMA2-70B to curate a large-scale dataset of ≈250K nostalgic English tweets

PUBLICATIONS

[1] **Stanley Jothiraj, Fiona Victoria**, Hutchinson Rebecca, Lauren Ponisio, Julia Jones, Nicole Martinez (2026). Plant-Pollinator Interactions over more than a Decade of Environmental Change. *under review*

[2] **Fiona Victoria Stanley Jothiraj**, Arunaggiri Pandian, Seth A. Eichmeyer. (2026) DECOR: Deep Embedding Clustering with Orientation Robustness. *AAAI 2026 Bridge Program on Knowledge-guided Machine Learning*. Also selected at the *AAAI 2026 Workshop on AI to Accelerate Science and Engineering (AI2ASE)*

[3] Shen Fang-Yu, **Stanley Jothiraj, Fiona Victoria**, Hutchinson Rebecca A, Hallman Tyler, Curtis Jenna R, and Robinson, W. Douglas. (2025). Species Distribution Model Performance Improves When Habitat Characterizations are Centered on Detected Individuals Instead of Observers. *Ecological Indicators*
[4] **Stanley Jothiraj, F. V.**, & Mashhadi, A. (2024). Phoenix: A Federated Generative Diffusion Model. *Companion Proceedings of the ACM on Web Conference 2024 (WWW 2024)*
[5] **Stanley Jothiraj, F. V.**, Hong L., & Mashhadi, A. (2024) Nostalgia on Twitter: Detection and Analysis of a Large-Scale Dataset. *Proceedings of the Association for Information Science and Technology (ASIS&T)*
[6] Fang-Yu Shen, **Fiona Victoria Stanley Jothiraj**, Tyler A. Hallman, Rebecca A. Hutchinson, W. Douglas Robinson. (2024). Does species distribution model performance improve when habitat sampling is centered on bird locations instead of surveyor locations? *American Ornithological Society Annual Meeting*
[7] **Stanley Jothiraj, F. V.** (2022). Time Series Prediction for Food Sustainability. *arXiv preprint*
[8] Shamsaddini, Vahid, **Stanley Jothiraj, Fiona Victoria**, Chen, Mandy, & Mashhadi, Afra. (2022). Empirical Dynamic Modelling of the Multi-Source Park Visitation Data. *Data for Policy Conference*

INVITED TALKS

• Ai4 Conference	Solo Speaker @ Edge AI & Tiny ML Technical Track	2026
• ICCV CVAM Workshop	Creativity and Privacy in Advertising & Marketing	2025
• Decentralized AI Hackathon @ Stanford	Federated Learning for Generative Models	2025
• University of Washington	Federated Learning for Generative Diffusion Models	2025
• Flower Labs	Phoenix: The First Federated Generative Diffusion Model	2023

EXPERIENCE

Graduate Teaching Assistant Jan 2026 - June 2026
Oregon State University
- Courses: CS 464 Open Source Software & CS 493 Cloud Application Development
- Supported over 300 students in mastering open source workflows and cloud-native development practices
- Delivered feedback on assignments and led office hours to reinforce hands-on technical concepts

Data Scientist June 2023 - Oct 2023
Harvard in Tech
- Led NLP research as a thought leader in the Call for Action (CFA) team to detect unreliable news articles
- Influenced executive leadership to shape the roadmap of the CFA Data Science team
- Built transformer-based models to classify news headlines using open-source and in-house curated data
- Presented findings on fake news detection, feature engineering, and state-of-the-art NLP approaches

Machine Learning Engineer June 2019 - Mar 2020
Multicoreware Inc, India
- Designed and deployed EV Quantization logic in TensorFlow GPU for quantization-aware training and Tensorflow Lite inference of Deep Neural Networks
- Shipped open-source product to production powering Synopsys DesignWare EV Processors
- Mentored peers on quantization concepts and EV TensorFlow workflows using C++, Python, and Intel Intrinsics

Machine Learning Intern Dec 2018 - May 2019
Multicoreware Inc, India
- Developed a custom CNN model for skin cancer detection by training on gigabytes of clinical image data
- Implemented the LipSync™ API in Intel's OpenVINO via a high-level C++ inference engine for 5x speedup
- Deployed the LipSync quality control tool for OTT streaming providers using deep learning
- Demonstrated LipSync™ technology at the National Association of Broadcasters Show (NAB 2019)

AWARDS

- **Professional Development Award -** Oregon State University 2024/2025
- **Micron Innovation Award -** Micron Technology 2024
- **Outstanding Scholars Program -** Oregon State University 2023
- **Virtual Scholarship -** Grace Hopper Women in Computing Celebration 2022
- **Academic Excellence** (Ranked 1/80) - Robotics and Automation Engineering Association 2019
- **Monarch of the Month -** Individual contribution to TensorFlow quantization at Multicoreware 2019

MEDIA

• Oregon State University	AI student leverages OSU's interdisciplinary strengths	2025
• Semiconductor Engineering	Grouping Complex Wafer Defect Patterns Into Meaningful Clusters	2025
• The Hindu	App for Plus Two Students	2015
• Business Standard	School Girl Develops App for TN Engg Aspirants	2015
• India.com	School Girl Develops Application for Tamil Nadu Engg Aspirants	2015

SERVICE

Reviewer - Conference & Workshop

- ICML Workshop on Technical AI Governance Research (TAIGR) 2026
- ICLR Workshop on Generative AI in Genomics: Barriers and Frontiers (Gen^2) 2026
- AAAI Workshop on Artificial Intelligence with Biased or Scarce Data (AIBSD) 2026
- AAAI International Conference on Web and Social Media (ICWSM) 2025
- ACL Workshop on DravidianLangTech 2025
- NeurIPS Workshop on Women in Machine Learning (WiML) 2024–2025

Reviewer - Journal

- ACM Transactions on Knowledge Discovery from Data (TKDD) 2025
- IEEE Transactions on Mobile Computing (TMC) 2024
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2024

CERTIFICATIONS

2022 Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning - Coursera
2022 Convolutional Neural Networks in TensorFlow - Coursera
2022 Natural Language Processing in TensorFlow - Coursera
2022 Introduction to Big Data - Coursera
2022 Taming Big Data with Apache Spark and Python – Udemy
2021 Amazon Web Services (AWS) - Machine Learning Specialty - Certified

SKILLS

- **Languages:** Python, R, C, C++, CUDA, MATLAB, SQL
- **A.I Tools:** TensorFlow, PyTorch, Diffusers, Transformers, OpenVINO, Keras, HuggingFace, Flower, Caffe, Scikit-learn, OpenCV, Kats, Pandas, NumPy, Jupyter, PySpark, Matplotlib, SciPy, Weights & Biases
- **R Libraries:** Dismo, Glmnet, Unmarked, DynamicSDM, Maxnet, Raster, Terra, Tidyverse, Nimble
- **Other Tools:** AWS (IoT Core, Sagemaker, Lambda, Kinesis, Glue, S3, SNS), Azure (IoT Hub, Stream Analytics, Function App), LaTeX, Git, Google Earth Engine, Docker